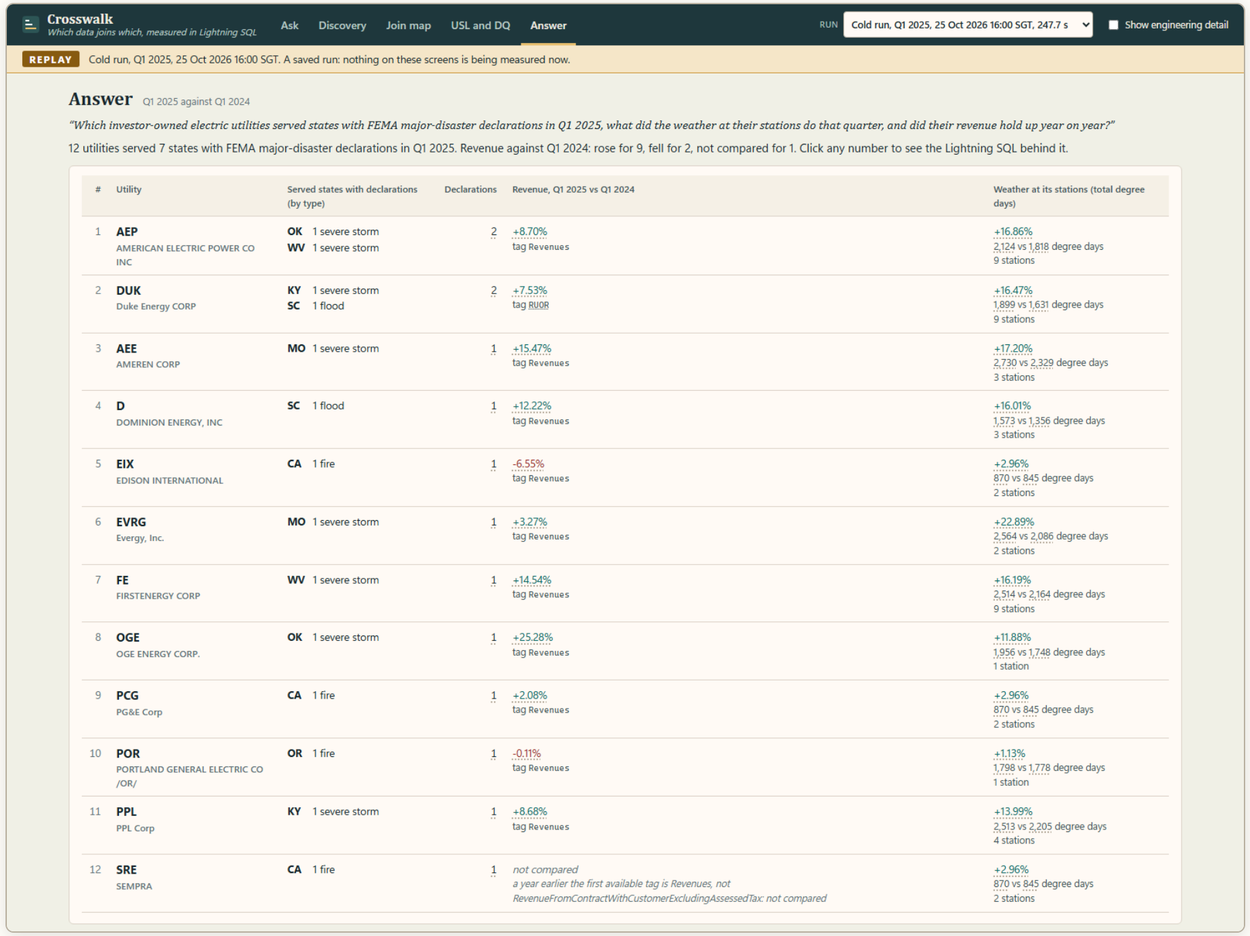
Crosswalk

A question that spans three US agencies, and no data engineer to join them. Crosswalk finds which public datasets join which, measures every join key in Lightning SQL with its orphans, builds a governed semantic layer from the keys that hold, and answers with the SQL behind every number.

Zetaris	The only data path: OpenFEMA, SEC EDGAR, NOAA GHCN-Daily and company_dns registered as Lightning tables, cached, and joined in Lightning SQL; a USL with primary and foreign keys and data-quality rules; the SQL text is the citation.
Meterless	H-MEM remembers the keys that held; a Markovian chain carries a small fixed state between chunks; the World Model records every measured key and since when.
NVIDIA	Nemotron 3 Nano 4B through local Ollama chooses the next step the checklist allows and proposes join keys from a menu; every proposal is measured before use. 0 cloud model calls.



Crosswalk — *Which data joins which, measured in Lightning SQL* | Ask | Discovery | Join map | USL and DQ | Answer | RUN: Cold run, Q1 2025, 25 Oct 2026 16:00 SGT, 247.7 s | Show engineering detail

REPLAY — Cold run, Q1 2025, 25 Oct 2026 16:00 SGT. A saved run: nothing on these screens is being measured now.

Answer Q1 2025 against Q1 2024

"Which investor-owned electric utilities served states with FEMA major-disaster declarations in Q1 2025, what did the weather at their stations do that quarter, and did their revenue hold up year on year?"

12 utilities served 7 states with FEMA major-disaster declarations in Q1 2025. Revenue against Q1 2024: rose for 9, fell for 2, not compared for 1. Click any number to see the Lightning SQL behind it.

#	Utility	Served states with declarations (by type)	Declarations	Revenue, Q1 2025 vs Q1 2024	Weather at its stations (total degree days)
1	**AEP** AMERICAN ELECTRIC POWER CO INC	**OK** 1 severe storm **WV** 1 severe storm	2	+8.70% tag Revenues	+16.86% 2,124 vs 1,818 degree days 9 stations
2	**DUK** Duke Energy CORP	**KY** 1 severe storm **SC** 1 flood	2	+7.53% tag RUOR	+16.47% 1,899 vs 1,631 degree days 9 stations
3	**AEE** AMEREN CORP	**MO** 1 severe storm	1	+15.47% tag Revenues	+17.20% 2,730 vs 2,329 degree days 3 stations
4	**D** DOMINION ENERGY, INC	**SC** 1 flood	1	+12.22% tag Revenues	+16.01% 1,573 vs 1,356 degree days 3 stations
5	**EIX** EDISON INTERNATIONAL	**CA** 1 fire	1	-6.55% tag Revenues	+2.96% 870 vs 845 degree days 2 stations
6	**EVRG** Evergy, Inc.	**MO** 1 severe storm	1	+3.27% tag Revenues	+22.89% 2,564 vs 2,086 degree days 2 stations
7	**FE** FIRSTENERGY CORP	**WV** 1 severe storm	1	+14.54% tag Revenues	+16.19% 2,514 vs 2,164 degree days 9 stations
8	**OGE** OGE ENERGY CORP.	**OK** 1 severe storm	1	+25.28% tag Revenues	+11.88% 1,956 vs 1,748 degree days 1 station
9	**PCG** PG&E Corp	**CA** 1 fire	1	+2.08% tag Revenues	+2.96% 870 vs 845 degree days 2 stations
10	**POR** PORTLAND GENERAL ELECTRIC CO /OR/	**OR** 1 fire	1	-0.11% tag Revenues	+1.13% 1,798 vs 1,778 degree days 1 station
11	**PPL** PPL Corp	**KY** 1 severe storm	1	+8.68% tag Revenues	+13.99% 2,513 vs 2,205 degree days 4 stations
12	**SRE** SEMPRA	**CA** 1 fire	1	*not compared* *a year earlier the first available tag is Revenues, not RevenueFromContractWithCustomerExcludingAssessedTax: not compared*	+2.96% 870 vs 845 degree days 2 stations

Q1 2025: FEMA declared major disasters in 8 states, and 12 investor-owned utilities served 7 of them. Against Q1 2024 their revenue rose for 9, fell for 2 and was not comparable for 1; degree days rose at the stations of all 12.

Umar Mohktar, Singapore - solo

Three agencies publish the pieces; no column joins them as published

The user

An analyst at a research desk, a state commission, a newsroom or a small fund who is handed a cross-agency question and has no data engineer on call.

The question

> *Which investor-owned electric utilities served states with FEMA major-disaster declarations in Q1 2025, what did the weather at their stations do that quarter, and did their revenue hold up year on year?*

Why the join is hard (each measured)

- SEC frames carry the company number as an integer, the filer profiles as a 10-digit string: joined as published, **0 of 43** utilities match; padded to 10 digits, 42 of 43.
- FEMA rows are designated counties (116 rows for 10 declarations in 8 states); utilities serve states, curated from each utility's 10-K.
- A weather station's state sits in columns 39-40 of a fixed-width text file of 133,039 lines.
- Alaska has a declaration and no seed utility; MGE Energy's SIC 4900 is missing from the SIC reference.

FEMA (OpenFEMA)

Disaster Declarations Summaries v2: one row per declaration and designated county, with the state as two letters and as a FIPS code. Major-disaster (DR) declarations dated in the quarter.

SEC EDGAR (XBRL)

Frames: one revenue fact per filer and quarter under a tag, company number unpadded, location as `US-XX`. Submissions: the filer profile with the 10-digit company number, SIC code and business-address state.

NOAA GHCN-Daily

Daily TMAX and TMIN per station (parquet by year on the AWS Open Data bucket) and the station inventory `ghcnd-stations.txt`, fixed-width text with the state inside each line.

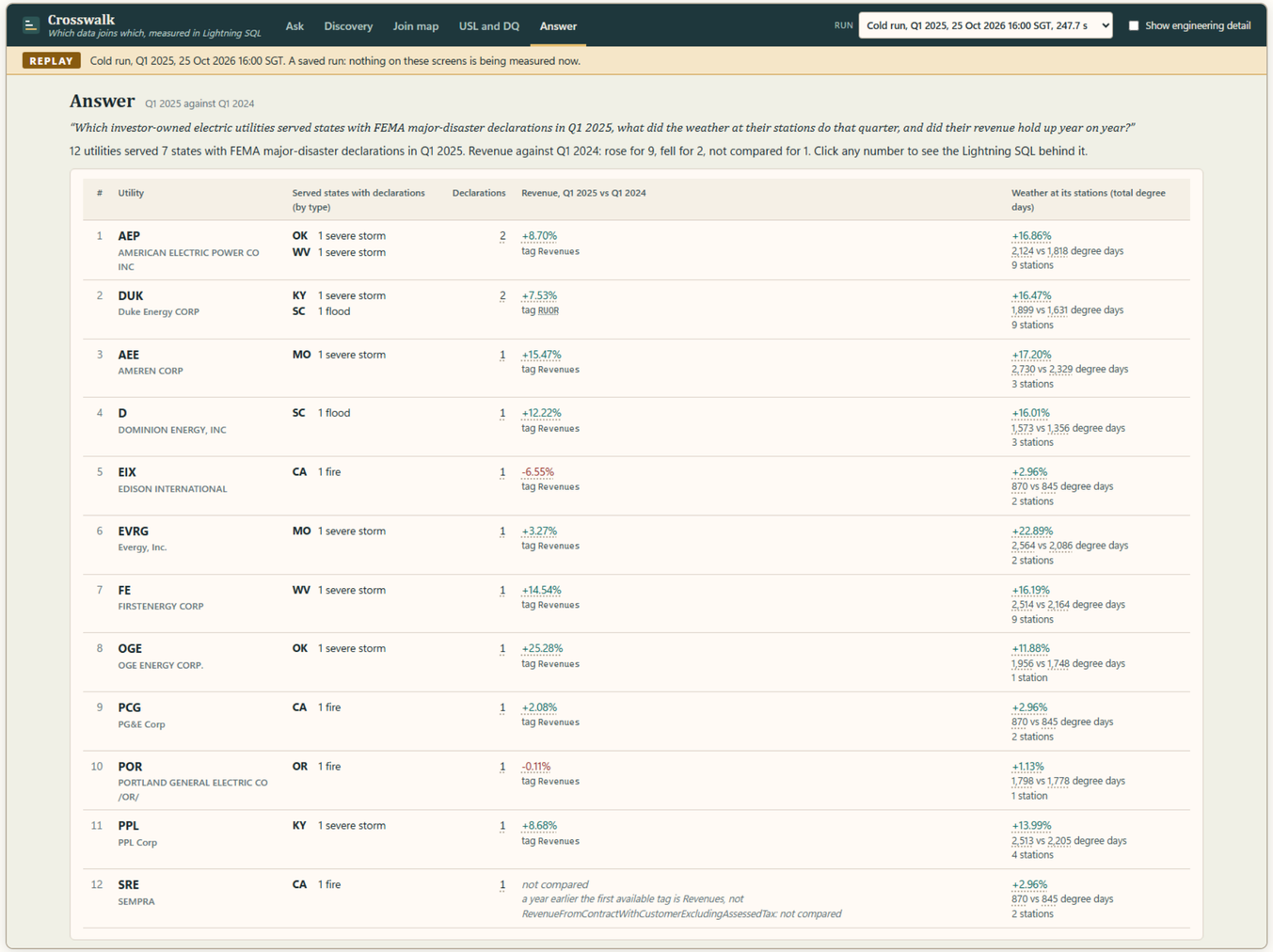
Two references

The company_dns SIC list (1,005 codes) and a state reference (50 states and DC with FIPS codes). Seeds: 43 utilities with the states they serve and 56 first-order weather stations, curated from public filings and labelled curated.

Counts from zetaris/README.md (measured 25 Oct 2026) and docs/eval-report.md section 4 (the five keys by hand through the gateway).

THE ANSWER

12 utilities served 7 declared states; every number opens its SQL



Crosswalk
Which data joins which, measured in Lightning SQL

Ask | Discovery | Join map | USL and DQ | Answer

RUN: Cold run, Q1 2025, 25 Oct 2026 16:00 SGT, 247.7 s | Show engineering detail

REPLAY Cold run, Q1 2025, 25 Oct 2026 16:00 SGT. A saved run: nothing on these screens is being measured now.

Answer Q1 2025 against Q1 2024

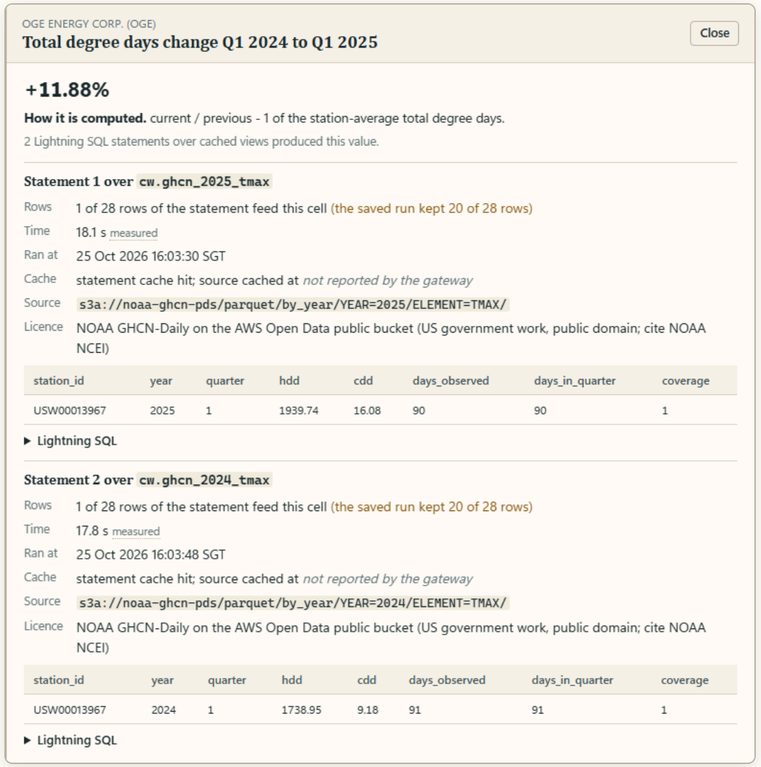
"Which investor-owned electric utilities served states with FEMA major-disaster declarations in Q1 2025, what did the weather at their stations do that quarter, and did their revenue hold up year on year?"

12 utilities served 7 states with FEMA major-disaster declarations in Q1 2025. Revenue against Q1 2024: rose for 9, fell for 2, not compared for 1. Click any number to see the Lightning SQL behind it.

#	Utility	Served states with declarations (by type)	Declarations	Revenue, Q1 2025 vs Q1 2024	Weather at its stations (total degree days)
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Saved cold run, 25 Oct 2026 16:00 SGT (REPLAY). The FEMA join selects the utilities; none is hand-picked. Revenue levels are hidden here: the repository keeps only the changes.

FEMA declared major disasters in 8 states in Q1 2025; 12 of the 43 seed utilities serve 7 of them (no seed utility serves Alaska). Revenue against Q1 2024 rose for 9, fell for EIX (−6.55%) and POR (−0.11%), and is not compared for SRE, whose first available tag differs between the two quarters.



OGE ENERGY CORP. (OGE)

Total degree days change Q1 2024 to Q1 2025

Close

+11.88%

How it is computed. current / previous - 1 of the station-average total degree days.

2 Lightning SQL statements over cached views produced this value.

Statement 1 over `cw.ghcn_2025_tmax`

Rows 1 of 28 rows of the statement feed this cell (the saved run kept 20 of 28 rows)
Time 18.1 s measured
Ran at 25 Oct 2026 16:03:30 SGT
Cache statement cache hit; source cached at *not reported by the gateway*
Source `s3://noaa-ghcn-pds/parquet/by_year/YEAR=2025/ELEMENT=TMAX/`
Licence NOAA GHCN-Daily on the AWS Open Data public bucket (US government work, public domain; cite NOAA NCEI)

station_id	year	quarter	hdd	cdd	days_observed	days_in_quarter	coverage
USW00013967	2025	1	1939.74	16.08	90	90	1

▸ Lightning SQL

Statement 2 over `cw.ghcn_2024_tmax`

Rows 1 of 28 rows of the statement feed this cell (the saved run kept 20 of 28 rows)
Time 17.8 s measured
Ran at 25 Oct 2026 16:03:48 SGT
Cache statement cache hit; source cached at *not reported by the gateway*
Source `s3://noaa-ghcn-pds/parquet/by_year/YEAR=2024/ELEMENT=TMAX/`
Licence NOAA GHCN-Daily on the AWS Open Data public bucket (US government work, public domain; cite NOAA NCEI)

station_id	year	quarter	hdd	cdd	days_observed	days_in_quarter	coverage
USW00013967	2024	1	1738.95	9.18	91	91	1

▸ Lightning SQL

The drawer behind OGE's +11.88%: two Lightning SQL statements over the cached GHCN views (18.1 s and 17.8 s), the row each returned, source and licence.

Answer identical in every field across 6 cold Day-2 runs (docs/eval-report.md section 1). Weather is the total degree days (heating + cooling) averaged over each utility's seed stations.

Six discovery tools in a fixed checklist; the model's key proposals are scored

`search_catalogue`
the Zetaris getting-started manifest, the registered views and the OpenFEMA dataset catalogue (a cached view)

→

`probe_source`
DESCRIBE, five rows and distinct counts of candidate key columns

→

`propose_keys`
up to six {left, right, transform} from a menu of probed columns and a closed transform set

→

`measure_key`
match rate in Lightning SQL over cached views; orphans listed

→

`register_source`
catalogue sources only, under a signed contract

→

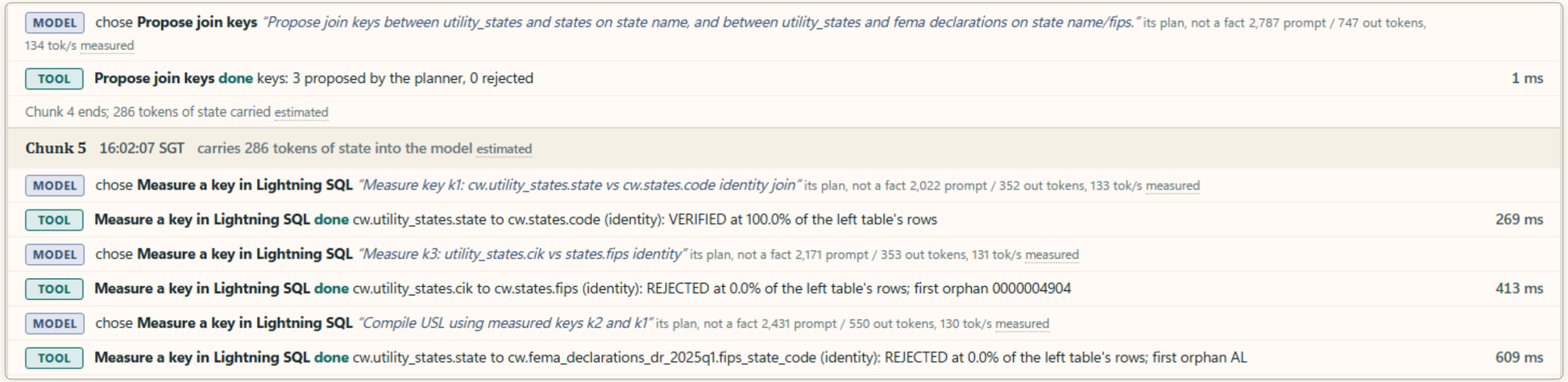
`compile_usl`
the plan from the verified keys, diffed against the governed USL; two DQ rules live

Model against a deterministic baseline, same probes, same SQL (5 cold runs)

Proposer	Made	VERIFIED	PARTIAL	REJECTED	Unmeasurable
Nemotron 3 Nano 4B, choosing from a menu of probed columns	11	3	1	7	0
Name-and-statistics baseline (no model)	30	25	0	5	0

The baseline wins. The model is kept as a second proposer whose every proposal is measured in Lightning SQL before anything uses it. It found no key the baseline missed: each key it got VERIFIED was also measured from a baseline proposal in the same run.

Then `answer` (fixed SQL templates), `not_tested` and `finalize` (the note, checked word by word against the join map and the answer). The checklist (sources found, shapes probed, keys proposed, keys measured, sources registered, USL built, DQ run, answered, not-tested listed, note written) fixes the order; the model chooses among the steps it allows and proposes keys. It never writes SQL or DDL and never computes a number. A reply that is invalid twice hands the step to the host.



Discovery timeline, saved cold run 25 Oct 2026 16:00 SGT: the model's three proposals, then each measured in Lightning SQL: 1 VERIFIED, 2 REJECTED at 0.0% of the left table's rows.

Model and baseline counts: docs/eval-report.md section 2 (cold runs with the model on Day 2). Transforms: identity, upper, lpad10, substr, strip_prefix('US-'), yyyymmdd_to_date, iso_to_quarter.

THE JOIN MAP

Every key measured with both rates and its orphans; the no's stay on the map

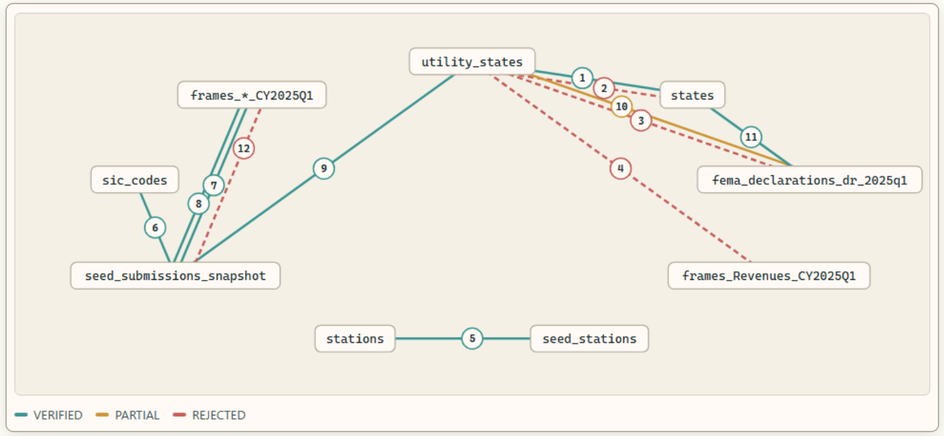
Key (left → right)	Transform	Rows matched	Distinct values	Verdict	Orphans
(a) submissions company number → frames company number	`lpad10`	42 / 43 (97.7%)	42 / 43	VERIFIED	IDACORP (no Q1 2025 fact)
(a) control: the same, without padding	`identity`	0 / 43 (0.0%)	0 / 43	REJECTED	all 43
(b) frames location → business-address state	`substr(4,2)`	42 / 42 (100%)	42 / 42	VERIFIED	none
(c) FEMA state → states the utilities serve	`identity`	7 / 8 (87.5%)	7 / 8	PARTIAL	AK
(c) FEMA FIPS state code → state reference	`identity`	8 / 8 (100%)	8 / 8	VERIFIED	none
(d) seed station id → GHCN station inventory	`identity`	56 / 56 (100%)	56 / 56	VERIFIED	none
(e) utility SIC → SIC reference	`identity`	42 / 43 (97.7%)	3 / 4 (75.0%)	VERIFIED	MGE Energy (4900)
(c) in the warm run, Q1 2024: FEMA state → served states	`identity`	8 / 11 (72.7%)	8 / 11	REJECTED	AK, ME, VT

VERIFIED — At least 95% of the left table's rows found a match on the right under this transform. Safe to join on; any rows that did not match are listed as orphans.

PARTIAL — Between 80% and 95% matched. Usable with care: the orphans listed drop out of any join on this key.

REJECTED — Fewer than 80% matched. This key does not join the two tables with this transform.

Rate — Matched rows over the rows of the left table; the verdict is read from it. The distinct-value reading is shown beside it and never decides (it would misread SIC, where 43 utilities share 4 codes).



Join graph of the cold run: 12 keys measured, 7 VERIFIED, 1 PARTIAL, 4 REJECTED, each one Lightning SQL statement over cached views.

A VERIFIED rate measures containment, not meaning

Two lists of US state codes overlap, so `hq_state` → station state (28 of 28) and served state → station state (43 of 43) also measure VERIFIED. Nobody would join on them; the join map shows them beside the governed keys without saying so.

Cold run 25 Oct 2026 16:00 SGT and warm run 15:56 SGT (Join map screen and run files); the same numbers by hand in docs/eval-report.md section 4. Left = the table whose rows are counted.

ARCHITECTURE

One data path through Zetaris; the model only chooses and proposes

Analyst

Browser UI on 127.0.0.1:8804 (Ask, Discovery, Join map, USL and DQ, Answer; telemetry footer; REPLAY or LIVE, never both) or the CLI: `npm run ask`, `npm run onboard`

↓ HTTP and server-sent events

Agent: Node 22 + TypeScript, no runtime dependencies

Markovian chain: at most 12 chunks, an 800-token carryover between them — METERLESS

H-MEM: key_verified and source_access memories with Zetaris provenance — METERLESS

World Model: datasets, columns, keys (`crosswalks_to`, valid from the measurement), utilities, states, declarations — METERLESS

Deterministic tools: the six discovery tools, answer templates, not-tested list, the note's word verifier — HOST

Ollama, 127.0.0.1:11434

local only; 0 cloud model calls

Nemotron 3 Nano 4B (Q4_K_M): picks the next checklist step, proposes keys from an enumerated menu (JSON schema), drafts the note — NVIDIA

↓ template id + parameters (the agent never sends SQL)

Gateway: Python 3.12, FastAPI, 127.0.0.1:8803

named Lightning SQL templates with a parameter allowlist (object names must be in the manifest); registrar plans under a signed contract; every statement logged with its time

↓ Lightning SQL over JDBC

Zetaris Lightning 2.4.3.1

CW_FEMA, CW_SEC, CW_DNS: OpenFEMA, SEC EDGAR frames and submissions, company_dns — REST

CW_GHCN: GHCN-Daily parquet 2024 and 2025, the station inventory as one-column CSV — FILESTORE

crosswalk_usl: 7 tables, 6 foreign keys with a DQ rule each, activated from the cached views — USL

Question path: cached SchemaStore views in container `cw` only; 0 of 399 statements over 20 s, slowest 18.1 s (the weather statement). **Onboarding path** (`npm run onboard`): register, cache, compile the USL, run every DQ rule bounded at 900 s.

↓ HTTPS, read by Zetaris only

Public sources: OpenFEMA API (fema.gov), SEC EDGAR (data.sec.gov, declared User-Agent, a fair-access limit of 10 requests a second), NOAA GHCN-Daily on the AWS Open Data bucket, company_dns; the Zetaris getting-started manifest is read at boot for the catalogue.

At runtime the agent never calls OpenFEMA, SEC EDGAR, NOAA or company_dns directly; only Zetaris does. Statement times: docs/eval-report.md section 6 (7 Day-2 runs).

Zetaris: registered sources, cached views, a governed USL, SQL as the citation

Lightning database	Source, as registered	Counted against the source (25 Oct 2026)
CW_FEMA (REST)	Declarations, DR filter percent-encoded; the OpenFEMA dataset catalogue	116 rows, 10 declarations, 8 states (Q1 2025); 72 / 14 / 11 (Q1 2024)
CW_SEC (REST)	Frames for 3 revenue tags × 2 quarters; 43 filer profiles	20 to 2,654 facts per frame, one row per filer
CW_DNS (REST)	company_dns SIC list	1,005 codes
CW_GHCN (filestore)	Station inventory (CSV, | delimiter, parsed by SUBSTR); parquet TMAX and TMIN, 2024 and 2025	133,039 stations; about 20,800 rows per year and element for 57 stations

Data-quality rules one rule per foreign key: every value must exist in the table it points to

Rule (foreign key)	At onboarding, 25 Oct 2026 13:46:21 SGT			Live in this run, 25 Oct 2026 16:03:19 SGT		
	valid	invalid	time	valid	invalid	time
`utility.sic` to `sic_code.sic`	42 of 43	1	135.5 s			*not run*
`utility_state.cik` to `utility.cik`	97 of 97	0	72.0 s			*not run*
`utility_state.state` to `state.code` IN THIS RUN	97 of 97	0	885 ms	97 of 97	0	759 ms
`fema_declaration.state` to `state.code` IN THIS RUN	116 of 116	0	2.9 s	116 of 116	0	4.8 s
`station.state` to `state.code`	56 of 56	0	37.7 s			*not run*
`revenue_quarter.cik` to `utility.cik`	133 of 133	0	84.5 s			*not run*

At onboarding — Run by the onboarding command (npm run onboard): it confirms the sources and the governed USL, then runs every DQ rule one at a time (each bounded at 900 s). The screen reads the saved results; no question re-runs them. Read from the saved results of 25 Oct 2026 13:46:21 SGT: 6 rules, 1 invalid row.

Live in this run — Run live inside a question by compile_usl: only the two rules that finish in a few seconds. 2 rules ran at 25 Oct 2026 16:03:19 SGT.

Invalid — A row whose foreign-key value has no match in the referenced table (for example a SIC code missing from the SIC reference).

Invalid rows, in plain words

MGE ENERGY INC (MGEE) has sic 4900, and the sic_code table has no row with that value, so the rule "utility.sic must exist in sic_code.sic" marks this row invalid. The sic_code table: SIC code reference from company_dns V4. Electric utilities are 4911, 4931 and 4932. The row stays in the model and is named here; it is not dropped or patched.

cik	name	ticker	sic
0001161728	MGE ENERGY INC	MGEE	4900

Rule `utility.sic`, found at onboarding.

Governed USL. `crosswalk_usl`: 7 tables with primary keys and 6 foreign keys, each a key measured before it was declared; activated from the cached views, because a raw REST table re-fetches its endpoint on every read. Each question runs 2 DQ rules live; the other 4 are read from the onboarding run of 25 Oct 13:46 SGT, where MGE Energy is the one invalid row.

Registration at run time. `register_source` creates, caches and counts a catalogue source under a signed contract. A Day-1 warm run registered three Q1 2023 revenue frames live, about 1 s each; the warm run recorded for the video registered FEMA Q1 2023 in 1.6 s and three Q1 2022 frames. In the Day-2 evaluation runs every call answered already registered.



OGE ENERGY CORP. (OGE)

Revenue change Q1 2024 to Q1 2025 Close

+25.28%

How it is computed. current / previous - 1, computed only when the first available tag is the same in both quarters.

2 Lightning SQL statements over cached views produced this value.

Statement 1 over `cw.frames_Revenues_CY2025Q1`

Rows 1 of 12 rows of the statement feed this cell
Time 475 ms measured
Ran at 25 Oct 2026 16:03:29 SGT
Cache statement cache hit; source cached at *not reported by the gateway*
Source https://data.sec.gov/api/xbrl/frames/us-gaap/Revenues/USD/CY2025Q1.json
Licence SEC EDGAR (US government work, public domain); fair-access policy: at most 10 requests per second with a declared User-Agent; queried live, never redistributed

cik	xbrl_tag	val	accn
0001021635	Revenues	*hidden here*	0001021635-26-000012

▼ Lightning SQL

```
SELECT cik, xbrl_tag, val, accn FROM (SELECT cik, xbrl_tag, val, accn, ROW_NUMBER() OVER (PARTITION BY cik ORDER BY pri) AS rn FROM (SELECT LPAD(CAST(cik AS STRING), 10, '0') AS cik, 'Revenues' AS xbrl_tag, CAST(val AS DOUBLE) AS val, accn, 1 AS pri FROM cw.frames_Revenues_CY2025Q1 UNION ALL SELECT LPAD(CAST(cik AS STRING), 10, '0') AS cik, 'RevenueFromContractWithCustomerExcludingAssessedTax' AS xbrl_tag, CAST(val AS DOUBLE) AS val, accn, 2 AS pri FROM cw.frames_RevenueFromContractWithCustomerExcludingAssessedTax_CY2025Q1 UNION ALL SELECT LPAD(CAST(cik AS STRING), 10, '0') AS cik, 'RegulatedAndUnregulatedOperatingRevenue' AS xbrl_tag, CAST(val AS DOUBLE) AS val, accn, 3 AS pri FROM cw.frames_RegulatedAndUnregulatedOperatingRevenue_CY2025Q1) u WHERE u.cik IN ('0000004904', '0001326160', '0001002910', '0000715957', '0000827052', '0001711269', '0001031296', '0001021635', '0001004980', '0000784977', '0000922224', '0001032208')) r WHERE rn = 1 ORDER BY cik
```

Statement 2 over `cw.frames_Revenues_CY2024Q1`

Rows 1 of 12 rows of the statement feed this cell
Time 440 ms measured
Ran at 25 Oct 2026 16:03:29 SGT
Cache statement cache hit; source cached at *not reported by the gateway*
Source https://data.sec.gov/api/xbrl/frames/us-gaap/Revenues/USD/CY2024Q1.json
Licence SEC EDGAR (US government work, public domain); fair-access policy: at most 10 requests per second with a declared User-Agent; queried live, never redistributed

cik	xbrl_tag	val	accn
0001021635	Revenues	*hidden here*	0000950170-25-060389

▶ Lightning SQL

The citation: OGE's +25.28% opens two Lightning SQL statements over the cached frames views. The revenue level is hidden in this image only.

Counts and timings from zetaris/README.md; live and onboarding DQ from the USL and DQ screen of the cold run 16:00 SGT; registrations from the Day-1 run record and docs/video/shotlist.md.

Meterless: remembered keys, a flat carried state, keys dated in the world

H-MEM: the keys that held

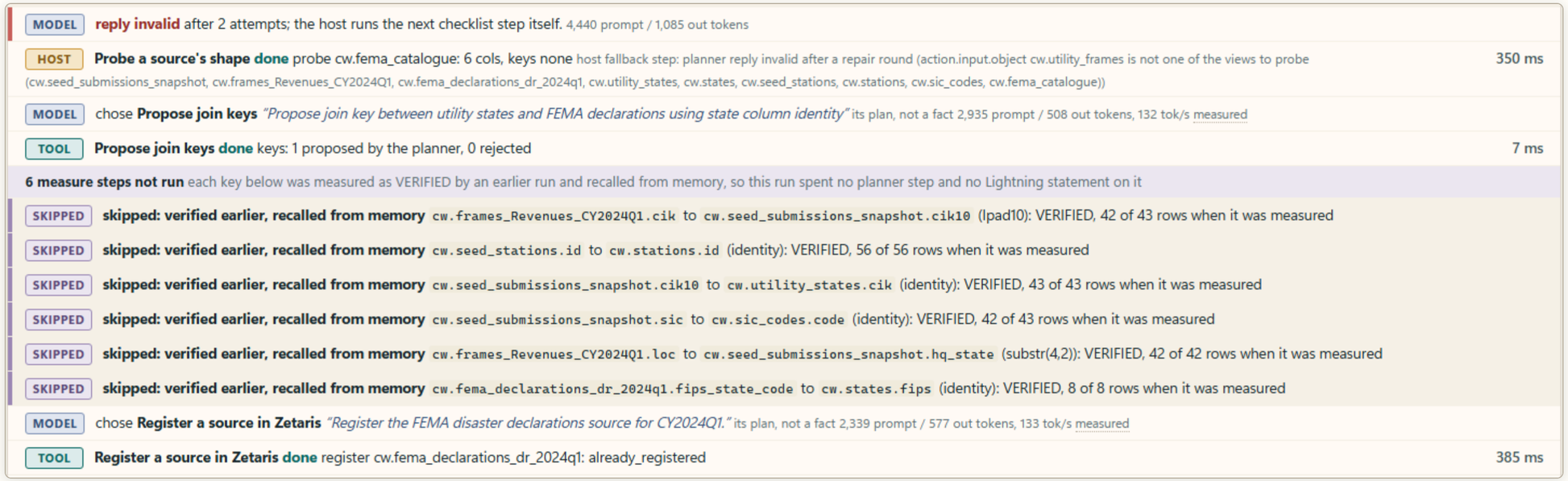
Every key measured VERIFIED is kept as a `key_verified` memory with its Zetaris provenance (SQL, object, time). The warm question (Q1 2024) recalled 6 and spent no planner step and no statement on them: 194.1 s, against 247.7 to 285.8 s for the cold runs. Seeded at boot with 46 memories: 23 from the Zetaris getting-started manifest, 10 from its known-to-fail notes, 12 source findings, 1 from the catalogue's known-to-fail list.

Markovian: a flat carried state

Cold run 16:00 SGT: 10 chunks of at most 12. The state carried between chunks was 273 to 451 tokens (estimated) against an 800-token cap. The first prompt of each chunk, as Ollama counted it, stayed between 1,782 and 2,236 tokens, while a prompt carrying the whole history would have reached 9,308 by chunk 10 (estimated).

World Model: since when

After the cold run: 444 events; 12 datasets, 71 columns and 25 `crosswalks_to` edges, one per measured key, carrying transform, rate and verdict, valid from the time Lightning measured it. Asked about OGE, it answers: DR-4862 (severe storms) declared in OK, valid from 2025-03-18; OGE serves OK (10-K seed, curated).



Warm run, Q1 2024, 25 Oct 2026 15:56 SGT: the six skipped measure steps, each recalled from a key memory with the rows it matched when it was measured.

Run files of the cold run 16:00 SGT and the warm run 15:56 SGT; the H-MEM seed record; the World Model read through `GET /api/world/1021635`. Engines vendored unchanged.

Nemotron 3 Nano 4B runs locally at 133 tokens/s; here it adds time, not keys

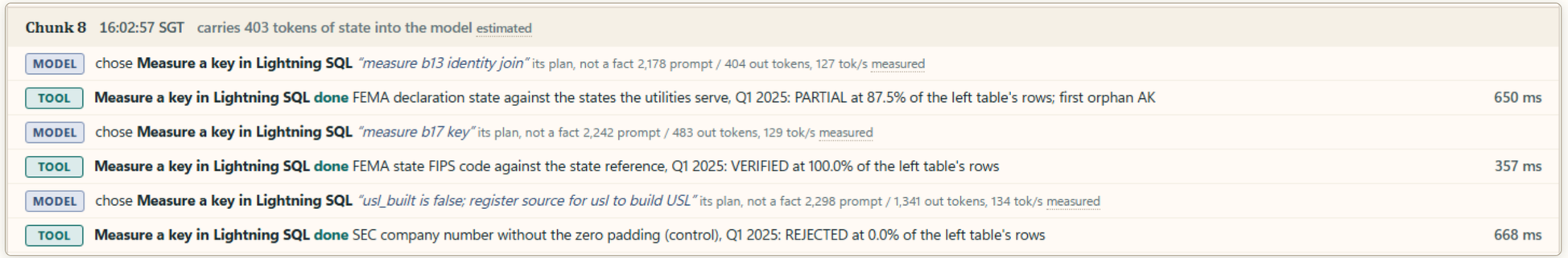
Cold question, same code	Wall	Chunks	Planner replies valid / calls (first try)	Host fallback steps	Output tokens	Median tok/s	Model keys VERIFIED / made	GPU util median	Answer
Think on	275.2 s	9	21 / 25 (19)	4	22,047	132.9	0 / 1	81%	reference
Think off	227.9 s	10	20 / 28 (19)	8	15,813	133.1	1 / 2	73%	identical
No model (host takes every step)	88.4 s	9	0 / 0	25	0	n/a	n/a	16%	identical

The model

`nemotron-3-nano:4b` through Ollama: 4.0B parameters, Q4_K_M, context 8,192, **2,881 MiB of VRAM**. JSON-schema replies: across the six runs with the model, 125 of 157 planner replies were valid; an empty reply cut by length is repaired once with thinking off. 0 cloud model calls.

What the numbers say, plainly

Without the model the same answer and governed keys arrive in 32% of the think-on wall time. One run per setting: the four think-on cold runs spread over 38.1 s, so the 47.3 s think-off gap is not claimed as an effect. The model was unloaded and reloaded mid-run 3 times (think on) and 2 times (off); the likely cause, a host-RAM guard, is not verified.



Cold run, chunk 8: each MODEL row is the model's stated plan ("its plan, not a fact"), each TOOL row what Lightning SQL measured: PARTIAL (AK), VERIFIED (FIPS), REJECTED (the control without padding).



Every screen carries the run's telemetry footer (cold run 16:00 SGT shown).

docs/eval-report.md section 5 and docs/eval/summary.json: GPU sampled every 500 ms by nvidia-smi; VRAM as Ollama reports it; tokens and tok/s are Ollama's counts.

EVALUATION

Reproducible, checked outside the agent, and compared with doing it by hand

Reproducibility

6 cold runs of the demo question (think on, off, no model): the same 12-row answer in every field, and the same numbers and orphans for the 7 governed join-map keys in every pair. The report and its JSON are regenerated from the saved runs by one command, with no gateway or model call.

Outside checks: numbers the agent did not compute

Check	Outside	Agent / Zetaris	Equal
OpenFEMA API, Q1 2025: rows / declarations / states	116 / 10 / 8	116 / 10 / 8	yes
OpenFEMA API, Q1 2024	72 / 14 / 11	72 / 14 / 11	yes
EVRG FY2024 10-K: states served	KS, MO	KS, MO	yes
PPL FY2024 10-K: states served	KY, PA, RI, VA	KY, PA, RI	no (VA has no declaration in either quarter)

Manual baseline: the five keys by hand, through the same gateway

	Statements	Decisions asked	Wall	Also produces
By hand (pre-written SQL)	27	24	179.5 s	the five keys and their orphans
Agent, cold, median	60	0	275.8 s	keys, cited answer, USL check, live DQ, checked note
Agent, warm (Q1 2024)	46	0	194.1 s	the same, verified keys recalled
Agent, cold, no model	56	0	88.4 s	same answer and keys, template note

An expert who already knows the five keys is faster by hand. The agent asks for none of the 24 decisions and adds what the hand session does not produce: every key measured with its orphans, the governed USL and its DQ, the cited answer, and memory for the next question. A person's time to write and read that SQL is not measured.

6 of 6
cold runs with the same answer in every field (think on, off, no model)

2 of 2
FEMA quarters where the OpenFEMA API equals the Zetaris cached view

24 → 0
decisions a person takes by hand to find the five keys, against one command

0 of 399
question-path statements over 20 s; the slowest, 18,135 ms, is the weather statement

docs/eval-report.md sections 1, 3, 4 and 6; docs/eval/summary.json, outside-checks.json, manual-baseline.json. The hand session ran gateway reader templates one at a time; 33.7 s of it was statement time.

What did not go well, and what the answer does not test

1. **The model loses the key-proposal comparison**
   11 proposals and 3 VERIFIED against the baseline's 30 and 25; every key it verified was also among the baseline's proposals. Some choices are pairs nobody would try (company number against FIPS code: 0 of 43); Lightning SQL rejects them, at the cost of a planner step and a statement each.

2. **Without the model, the same answer in about a third of the time**
   88.4 s with no model against 247.7 to 285.8 s with it, same answer and governed keys: on this question the planner and the note add time, not keys.

3. **The planner's replies and the note**
   125 of 157 planner replies valid; the host took 2 to 8 checklist steps itself per run. The model's note was replaced by the 250-word template in 3 of the 6 runs with the model; one early draft said revenue "held up" while EIX and POR fell, and the verifier rule was fixed afterwards.

4. **A VERIFIED rate is containment, not meaning**
   Two state-code lists overlap and measure VERIFIED (28 of 28, 43 of 43) though nobody would join on them.

5. **The curated seed has a gap; the station inventory is slow**
   PPL's 10-K also names Virginia, which the seed leaves out (no answer changes). Probing the 133,039-row station inventory costs about 27 s per cold run; each weather statement 18 s.

Not tested (on every answer)

- County geography: declarations are joined to utilities by state, not by the counties they name.
- Non-major declarations: emergency (EM) and fire-management (FM) declarations are left out.
- Fuel-cost pass-through: a revenue change is not a measure of disaster impact.
- Recasts: frames serve the latest filed value; restated comparatives are not reconciled.

Not measured

- A person's time for the manual baseline.
- More than one run per ablation setting; a second quantisation.
- Revenue against the companies' own reports; 41 of the 43 seed service-territory lists.
- Live registration in the Day-2 runs (all already registered), and the full DQ run on Day 2.

docs/eval-report.md "What did not go well" and "Not measured"; the Not-tested list as the Answer screen shows it.

NEXT, AND WHAT IT IS BUILT ON

Future work, data and code attributions

Future work

- Give the model sources whose keys the column names do not reveal, and keep scoring it against the baseline on the same probes.
- Flag VERIFIED keys that are only overlapping code lists.
- County-level geography: declaration counties against service territories.
- Like-for-like revenue with restatements, more quarters, the full DQ run on a schedule with its history.
- A timed manual baseline with analysts; repeated ablation runs and a second quantisation.

Pre-existing code, declared

- Meterless H-MEM, Markovian and World Model engines, vendored unchanged: 60 files, 13,189 lines (`git ls-files engines | xargs cat | wc -l`).
- `gateway/jdbc.py` (279 lines) adapted from the Zetaris getting-started repository.
- Scaffolding adapted from the same author's Weather-or-Not (github.com/GitHubMoreLikeGitGud/weather-or-not) and ClearShelf (github.com/GitHubMoreLikeGitGud/clearshelf): 26 code files carry that header, 8,523 lines today, rewritten in part (`git grep -l "Adapted from Weather-or-Not"` over agent, gateway, ui, scripts, zetaris, seeds).

Data

OpenFEMA: Disaster Declarations Summaries v2 and the dataset catalogue. Data from OpenFEMA (fema.gov); this product is not endorsed by FEMA.

SEC EDGAR: XBRL frames and submissions; US government work, public domain; queried live with a declared User-Agent, never redistributed.

NOAA GHCN-Daily: NOAA National Centers for Environmental Information, Global Historical Climatology Network daily, AWS Open Data bucket.

company_dns: SIC code list, Apache-2.0.

Zetaris getting-started manifest and known-to-fail notes (Apache-2.0) seed the catalogue and memories.

Developed in Cursor; details in docs/how-cursor-was-used.md. Seven pieces (C1-C7) were written by Cursor Agent from task files in docs/cursor-tasks, reviewed and committed as "(Cursor Agent)".

Code: github.com/GitHubMoreLikeGitGud/crosswalk (Apache-2.0). Model: NVIDIA Nemotron under the NVIDIA Open Model License.